

Course Project Requirements

Course: Neural Network Essentials

Deadline: 17 May 2026

1 Overview

The course project consists of writing an **extended abstract** (maximum 2 pages, excluding references) that demonstrates the student's ability to **understand, analyze, and apply concepts related to Neural Networks**.

The project is designed as a **mini research exercise**, encouraging students to explore a focused topic, implement or analyze a method, and clearly communicate their findings in a scientific format.

The structure is aligned with academic submission formats (e.g., short papers for workshops or summer schools).

2 Project Type

Each student must complete an **individual research-oriented project** in the field of Neural Networks.

Examples of suitable directions include (but are not limited to):

- Designing or improving a neural network architecture (e.g., CNNs, RNNs, Transformers, GNNs)
- Applying neural networks to a specific task (e.g., time series, NLP, computer vision, anomaly detection)
- Analyzing training dynamics, optimization, or generalization properties
- Investigating interpretability or explainability methods
- Exploring robustness, overfitting, or regularization techniques
- Comparing neural architectures or training strategies

The project does **not need to be fully mature or publishable**, but it must demonstrate:

- Clear problem formulation
- A well-defined methodology
- Experimental or analytical insights

3 Extended Abstract Format

- **Length:** maximum 2 pages (excluding references)
- **Font:** minimum 11pt
- **Format:** PDF

- **Recommended:** LaTeX

Suggested structure:

- Introduction
- Related Work (brief)
- Method / Approach
- Experiments / Analysis
- Results
- Conclusions
- References

Figures, tables, and diagrams are encouraged where relevant.

4 Evaluation Criteria

The project will be evaluated based on the following criteria:

- **Clarity of presentation** (structure, writing, visual elements)
- **Understanding of neural network concepts**
- **Technical depth and correctness**
- **Quality of experiments or analysis**
- **Interpretation of results**
- **Positioning within existing literature**
- **Bonus:** originality or novel insights

5 Individual Work

This project must be completed **individually**.

- Collaboration between students is **not allowed**
- Each submission must reflect the **student's own work and understanding**
- Any external contributions or resources must be **properly cited**

6 Use of LLM-based Assistants

The use of Large Language Models (LLMs) and AI assistants is **allowed**, under the following conditions:

- All presented content must be **understood, verified, and assumed by the student**
- Students must demonstrate **critical thinking** and not rely blindly on generated outputs
- Results should be **refined, validated, and improved** beyond initial generated responses

The goal is to assess:

- the ability to reason and validate information;
- the ability to build meaningful insights;
- the capacity to go beyond superficial answers.

Unverified or poorly understood content will be penalized.

7 Submission Details

- **Deadline:** 17 May 2026
- **Format:** PDF
- **File name:** <FirstName_LastName_NNE2026.pdf

We look forward to your projects and encourage you to explore ambitious ideas!